

Huang Tang

Principal Artificial Intelligence Engineer & AI Solution Architect

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Professional Summary

Principal Artificial Intelligence Engineer with over 12 years of experience at The MITRE Corporation, specializing in end-to-end product development and model-based systems engineering. Expert in Generative AI, including Large Language Models (LLMs), Retrieval Augmented Generation (RAG), and Knowledge Graphs. Proven track record of technical leadership in delivering tailored AI/ML solutions to federal sponsors including DISA, DHA, FAA, and DHS. Seeking to leverage deep expertise in Graph Neural Networks, Transformers, and MLOps to drive innovation.

Technical Skills

- **Generative AI & NLP:** Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Agentic AI, Transformers, Natural Language Processing (NLP), Question Answering and Recommendation Systems.
- **Machine Learning on Graphs:** Graph Neural Networks (GNNs), Causal Learning, Reinforcement Learning (RL), Knowledge Graphs.
- **Frameworks & Tools:** PyTorch, TensorFlow, MxNet, Ray, MLflow, ONNX.
- **Database Systems:** Weaviate, Pinecone, ActiveLoop, Neo4J, PostgreSQL, MongoDB.
- **Programming:** Python, Spark, Java, R, JavaScript, Matlab.
- **Communication Skills:** Drove sponsor engagement across all organizational levels by translating complex technical solutions into clear, actionable insights.

Professional Experience

REI Systems | AI Solution Architect | Dec 2025 - Present

As an AI Solution Architect of the CTO Office, leading the establishment of the company's AI foundation and providing SME level consulting to AI efforts across the company.

- **Agentic AI & Large Language Models in Deep Research and Active Data Catalog:**
 - Designing and implementing agentic AI systems for multimodal data governance and automated querying
 - Developing LLM-based agents to support FDA entry inspections
 - Building LLM agents for supply chain traceability and disruption impact prediction

The MITRE Corporation | Principal Artificial Intelligence Engineer | Sep 2012 – Jun 2025

Led cross-functional teams to architect and deliver generative AI and natural language solutions for diverse government sponsors.

- **GenAI & LLM Prototyping (RAG/GraphRAG Focus):**

- Led the development of a "Document to (Logic) Graph" system, utilizing LLMs to auto-generate architectures from large collections of Systems Engineering design documents.
- Served as Tech Leader for a Question Answering System leveraging Knowledge Graph-Indexed RAG for contract management.
- Developed a recommendation system for aviation incident root causes using LLMs and Knowledge Graphs.

- **NLP & Document Understanding:**

- Led NLP Topic Modeling efforts for safety report topic mining within the aviation domain.
- Participated the "Document Deep Understanding" initiative to perform legal analysis and deconflict aviation safety regulations using LLMs.

- **Advanced Machine Learning & Technical Leadership:**

- Acted as Project Lead for multi-modal, multi-variate time-series anomaly detection and root cause discovery using Temporal Fusion Transformers.
- Spearheaded large-scale network traffic engineering projects utilizing Graph Neural Networks (GNN) and Reinforcement Learning (RL).
- Developed and served as Chief Engineer for the MERIT (Medical Evaluation Readiness Information Toolset) project, applying predictive time-series analysis to generate 6 to 12 months disability outcome predictions based on Electronic Health Records (EHRs) and service histories.
- Led the development of Multi-agent Reinforcement Learning for airport ground traffic management.
- Proposed and led the development of data assimilation technology for uncertainty reduction in long horizon predictions.
- Proposed and led the complex systems architecture auto design and modernization paths auto discovery using constraint programming and reinforcement learning.

Metavi, LLC Co-Founder | Feb 2011 – Aug 2012

Modeled and visualized aviation traffic control data on top of weather information.

George Mason University Research Scientist | 2005 – 2010

Conducted researches on Numerical Weather Prediction (NWP) models and atmospheric transport models for Chemical, Biological, Radiological, and Nuclear (CBRN) materials.

Education

- Ph.D. (All but Dissertation), Aerospace Engineering and Mechanics, University of Minnesota (1998–2005)
- M.S., Civil Engineering, Hunan University (1998)
- B.S., Engineering Mechanics, Hunan University (1993)

Selected Publications

- On the Exploration of Temporal Fusion Transformers for Anomaly Detection with Multivariate Aviation Time-series Data. DASC 2024.
- Medical Evaluation Readiness Information Toolset (MERIT): Developing a Data-driven Decision Support Tool to Augment Complex Clinical Decisions. Military Medicine (Volume 188) 2023.
- Adaptable Graph Network for Traffic Analysis Applications. DASC 2022.
- Deep Reinforcement Learning Applied to Airport Surface Movement Planning. DASC 2019.
- Use Data Assimilation to Reduce Uncertainties in Ensemble Forecasting for Strategic Traffic Flow Management. 18th Conference on Aviation, range, and Aerospace Meteorology. 2016
- Agile Simulation Based Analysis of Acquisition. Summer Computer Simulation conference 2014.

CLEARANCE & ELIGIBILITY

- U.S Citizen
- DoD Secret Clearance since 2012